

# Research Design — Reframing plan

MA in Computational Social Science, UC3M — Fall 2026

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Drafted 3 September 2026

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## Research Design — Reframing plan for Fall 2026

Drafted 2026-09-03. **Course starts Tue Sep 15, 2026** (12 days out).

### The proposal in one paragraph

The course currently spends **2.5 of its 5 lectures on causal-inference machinery** (potential outcomes  $\rightarrow$  DAGs  $\rightarrow$  biases  $\rightarrow$  FE/DiD/RDD/IV  $\rightarrow$  the same methods again in more detail), and roughly **half a lecture on measurement and description**. For a cohort at the *start* of a CSS master's who will meet estimation again with the math in AQM, that allocation is backwards: the things that actually break MA theses are bad questions, found data of unknown provenance, and unvalidated measures — not a shaky grasp of the DiD estimator. The reframing **compresses causality to  $\sim$ 1.5 sessions, gives measurement + description a full session, and opens a new block on where computational data comes from** (found vs. designed data, algorithmic confounding, non-representativeness by construction, access). CSS stops being a source of example slides and becomes the course's design problem. Scope, session count, and slot times are unchanged.

### Files

| File             | What's in it                                                                          |
|------------------|---------------------------------------------------------------------------------------|
| 01_diagnosis.md  | Audit of the five existing decks — what works, what's duplicated, what's out of scope |
| 02_spine.md      | The organizing logic: describe/explain/predict + the five design questions            |
| 03_sessions.md   | Session-by-session keep / cut / add, with time budgets                                |
| 04_examples.md   | Worked examples and paper candidates, with what each one teaches                      |
| 05_readings.md   | Assigned readings per session + instructor background                                 |
| 06_assessment.md | Milestone ladder, assessment weights, workshop format                                 |
| 07_todo.md       | Triaged task list: before Sep 15 / during term / next year                            |

## Headline changes

1. **Session 2 is rebuilt** around data provenance (the CSS core), and measurement moves out of it.
2. **Session 3 becomes a full session on measurement and description** — currently squeezed into the back half of session 2.
3. **Prediction is promoted to a first-class research goal** alongside description and explanation. It is currently commented out in `3_causality/causality.tex` (six dead frames).
4. **Session 5's duplicate methods block is cut** (~65 frames re-teaching FE/DiD/RDD/IV with regression algebra — both redundant and outside the stated no-statistics scope). The freed time goes to generalization, credibility/reproducibility, and research ethics.
5. **A milestone ladder** replaces the current jump from week 4 to the workshop: five short, cheap-to-grade deliverables that assemble into the final essay.
6. **Decks get capped at ~55 frames** with the overflow moved to explicit appendix sections. Current decks run 86–110 frames for a 3h slot that also has to fit a paper discussion.

## Known instructor intents already folded in

- The note at the top of `syllabus_research_design.tex` — *raise the final essay to 50–60%, drop participation* — is implemented in `06_assessment.md` (essay 55%, participation dropped).
- `syllabus_research_design.tex` still says **Fall 2025** and Sep 16; `index.md` already has the 2026 dates. The syllabus needs updating either way — see `07_todo.md`.

## If there isn't time for all of this

`07_todo.md` has a **minimal-change variant** that keeps the existing session order and gets most of the CSS benefit from three inserted blocks and one deletion. That version is realistically doable in the 12 days available.

## Diagnosis: audit of the current five decks

Frame counts and PDF pages as of 2026-09-03.

| Deck                        | Frames | PDF pp. | Verdict                                       |
|-----------------------------|--------|---------|-----------------------------------------------|
| 1_introduction/             | 39     | 85      | Good bones, slightly over-long on process     |
| 2_basics_quantitative_data/ | 86     | 141     | Doing the work of two sessions                |
| 3_causality/                | 91     | 132     | Strong but over-length; prediction block dead |
| 4_causal_inference_methods/ | 58     | 85      | Best-calibrated deck in the course            |
| 5_advanced_topics/          | 110    | 148     | Two decks stapled together; ~60% duplicative  |

Each folder holds one .tex deck of the same name (introduction.tex, basics\_quant\_data.tex, causality.tex, causal\_inference.tex, advanced.tex).

### What is working and should be protected

- **The Challenger opening** (session 1). A genuinely excellent cold open: an explanatory question, an obvious-in-hindsight selection problem, no statistics required, images already in img/challenger\*.jpg. Keep verbatim.
- **The RQ ladder** (session 2): Topic > RQ > Theory refined into Previous evidence > argument > RQ > Hypotheses, then the honest version with the “anecdotal argument” step. This is the most useful twenty minutes in the course for students who have never written a design. Keep and expand.
- **“Good RQs” criteria with counterexamples** (answerable / relevant / feasible / narrow, with the three-way contrast of good vs. too broad vs. many-questions-in-one). Keep.
- **Concepts as rule-based vs. ideal types**, and the insistence that *operationalize*  $\neq$  *measure*. Rare to see taught explicitly; keep.
- **The democratic-backsliding measurement contrast** (objective indicators vs. V-Dem subjective coding). Perfect no-math measurement-validity lesson.
- **The “usual suspects” bias sequence** (confounding  $\rightarrow$  reverse causality  $\rightarrow$  selection  $\rightarrow$  collider  $\rightarrow$  post-treatment), with the collider animation and the closing “what should *not* be controlled for” recap. Strongest teaching block in the course.
- **The three-design-templates overview** in session 4. This is already the right idea; the reframing promotes it to the organizing frame for the whole session.
- **Snow’s cholera** and the **street-names DiD** (instructor’s own work). Both keep.

### Problems, in order of how much they cost

#### 1. Session 5 is two decks stapled together, and most of it is out of scope

advanced.tex runs 110 frames. The first ~45 are a coherent session (generalization, meta-analysis, external validity, placebo/robustness logic, synthetic control). The remaining ~65 — the Causal methods again, in detail section — re-teach controlling, matching, propensity scores, fixed effects, DiD, RDD, and IV, this time *with* the regression machinery: Controlling in regression, Fixed effects and regression ( $\times 3$ ), RDD and regression, How does IV work? ( $\times 7$ ), Two approaches to matching: propensity score ( $\times 2$ ).

Two problems at once. It duplicates session 4, and it breaks the course's own stated scope — the syllabus promises “we will not cover statistical techniques or advanced data analysis.” **Cut the whole section**; park it as an optional appendix deck for students continuing to AQM.

## ***2. Nothing in the course is about where computational data comes from***

The MA is in Computational Social Science. CSS currently enters only as *examples* — Barberá's Twitter ideal points, Blumenstock's phone metadata, Müller-Crepon's roads, the Guess et al. platform experiment. All good examples, but no session treats the design problems that are **specific** to computational data:

- found/observational digital traces vs. data you designed to collect
- the platform as part of the data-generating process (algorithmic confounding)
- non-representativeness by construction, and why it is a property of the design not the sample
- drift: a measure that silently stops meaning what it meant
- access and reproducibility in the post-API era
- a classifier's output used as a variable — validation, and error that is not random
- ethics and informational risk at scale

Every one of these is teachable with zero statistics and zero code. This is the largest gap and the main thing the reframing fixes.

## ***3. Prediction is commented out***

`causality.tex` contains six consecutive commented frames: About prediction (in the social sciences) and About prediction  $\times 5$ , plus Prediction and explanation. For a CSS cohort this is the wrong thing to have cut. Prediction is a distinct research goal with its own design logic and its own failure modes, and it is what half the field the students are entering actually does. Restore it, and promote description/explanation/prediction to the course's organizing triad.

## ***4. Measurement and description are crammed into the back of session 2***

`basics_quant_data.tex` covers, in one 3h slot that also has a paper discussion: research questions, theory generation, concepts, operationalization, three distinct measurement issues, proxies, latent variables, validation, missing data, sampling bias, unit of analysis, levels of explanation, univariate and bivariate description, and a worked example. That is 86 frames and two sessions of material. In practice something gets dropped every year, and it is almost certainly description — which is then never taught, despite being the goal of a large share of computational work.

## ***5. Deck lengths are not compatible with the slot***

A 3h session realistically holds ~100–110 minutes of lecture once you account for arrival, recap, a break, and 45 minutes of paper discussion. At 1.5–2 min/frame that is **55–70 frames maximum**, and less when frames are overlay-heavy. Three of five decks exceed that; two exceed it badly. The fix is not faster delivery, it is an explicit appendix convention.

## 6. No scaffolding between week 4 and the workshop

The final essay is 40% of the grade and is the first research design most of these students will ever have written. The only checkpoints are a Re-cap and final essay slide in session 4 and the workshop itself in session 6. Students arrive at the workshop having started thinking about their design that week. A milestone ladder fixes this at near-zero grading cost — see 06\_assessment.md.

## 7. Infrastructure: no presenter notes anywhere

`grep -c '\note{'` returns **0 for all five decks**, and there is no `beamer_notes_preamble.tex` in `slides/`. The project conventions in `CLAUDE.md` describe a two-wrapper notes pattern used elsewhere in these course materials; this course never adopted it. Worth adding as decks get rewritten, not as a standalone task.

## 8. Minor

- The Charlie Kirk / rise of political violence running example in session 2 is topical and effective but will date fast. Keep the slot, mark it as a **swappable current-event example** so it is obvious what to replace each year.
- Two off topic: Controlling frames in causality.tex drift into regression a session early. Compress to one conceptual slide.
- Key ingredients (session 1) — typo.
- Research process in detail (1)–(6) is six slides for something that lands in two or three.

## The spine: what the course is organized around

### Problem with the current spine

The current order — intro → elements of quantitative data → causality → causal inference methods → advanced topics — is a **topic list inherited from the causal inference curriculum**. It works, but it has two costs. Students can't tell where they are in an argument, because there isn't one; and “advanced topics” is a name for a leftovers session, which is why deck 5 grew to 110 frames.

### Proposed spine

Two layers. A **triad** introduced in session 1 and revisited every week, and a **sequence of five design questions** that maps one-to-one onto the five lectures.

#### Layer 1 — three research goals (the triad)

| Goal            | The question                                         | A CSS example to anchor it                    |
|-----------------|------------------------------------------------------|-----------------------------------------------|
| <b>Describe</b> | What is the case? How much, where, for whom?         | Chetty et al. social capital atlas            |
| <b>Explain</b>  | Does X cause Y, and through what?                    | Guess et al. Facebook feed experiment         |
| <b>Predict</b>  | Given what I observe, what happens next / elsewhere? | Blumenstock et al. wealth from phone metadata |

Why this earns its place as the spine:

- It is the **cleanest way to make “what is your question?” concrete**. Most bad student designs are a description question with a causal answer bolted on, or a causal question answered with a predictive design. Naming the three goals gives you vocabulary to diagnose that in ten seconds during the workshop.
- Each goal has **different success criteria** — coverage and validity for description, identification for explanation, out-of-sample performance for prediction — so it motivates everything that follows.
- It is **the actual fault line in CSS**, and the students will hit it in every other course in the program. Getting it in week 1 of a 1-year MA pays off downstream.
- It restores the commented-out prediction material with a reason to exist.

Return to the triad at the end of every session with one slide: *what does this week change about how you'd describe / explain / predict?*

#### Layer 2 — the five design questions

| # | Session            | The question the session answers                                                |
|---|--------------------|---------------------------------------------------------------------------------|
| 1 | Introduction       | <b>Why design at all?</b> What does it mean to answer a question with evidence? |
| 2 | Questions and data | <b>What is the question, and where would the data come from?</b>                |
| 3 | Measurement        | <b>Do the numbers mean what you think they mean?</b>                            |
| 4 | Causality          | <b>What comparison would license a causal claim?</b>                            |

| # | Session                 | The question the session answers                                                    |
|---|-------------------------|-------------------------------------------------------------------------------------|
| 5 | Designs and credibility | <b>What variation can you actually exploit, and how far does the answer travel?</b> |
| 6 | Workshop                | Students run their own designs through questions 1–5                                |

The wording matters: each is phrased as something a student must be able to answer about **their own project**, not as a topic to be covered. Put the five questions on a slide in session 1, and open each subsequent session with that slide, current question highlighted. It costs one slide a week and it is what makes a five-week course feel like an argument.

### What changes in the allocation

|                         | current      | proposed     |               |
|-------------------------|--------------|--------------|---------------|
| questions / theory      | ~0.5 session | ~0.5 session |               |
| data provenance (CSS)   | –            | ~0.5 session | <– new        |
| measurement + descr.    | ~0.5 session | ~1 session   | <– doubled    |
| causality               | ~1 session   | ~1 session   |               |
| design templates        | ~1 session   | ~0.5 session | <– compressed |
| methods again, detailed | ~0.7 session | –            | <– cut        |
| generalization          | ~0.3 session | ~0.3 session |               |
| credibility + ethics    | –            | ~0.4 session | <– new        |

The two additions are paid for almost entirely by the deletion in session 5, which is why the reframing does not require the course to lose anything it currently teaches well.

### The scope line, stated once so it can be enforced

The syllabus promises no statistical techniques and no code. The operational version of that promise, for every slide in the course:

Teach **what variation a design exploits and what has to be true for it to work**. Never teach how the corresponding quantity is estimated.

So: DiD is “compare the change in the treated group to the change in a group that should have moved in parallel, and the design lives or dies on whether that parallel movement is credible.” Not a two-way fixed effects specification. This line is what session 5’s detailed methods block crosses, and having it written down makes the cut obvious rather than arbitrary.

## Session-by-session plan

Slot is 18h–21h (session 6 is 15h–21h). Working budget per lecture session:

|             |                                         |                          |
|-------------|-----------------------------------------|--------------------------|
| 18:00–18:15 | arrival, recap slide, milestone hand-in |                          |
| 18:15–19:05 | lecture block A                         | (~50 min, ~25–30 frames) |
| 19:05–19:15 | break                                   |                          |
| 19:15–20:05 | lecture block B                         | (~50 min, ~25–30 frames) |
| 20:05–20:55 | paper discussion                        | (~50 min)                |
| 20:55–21:00 | next week + milestone brief             |                          |

**Target: ~55 frames in the main body.** Anything else goes after a \section{Appendix} and is skipped live.

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### Session 1 — Why design at all? (Tue Sep 15)

Source deck: 1\_introduction/introduction.tex (39 frames — closest to right already).

#### *Keep*

- Challenger cold open, in full. Best thing in the course.
- Types of research (theoretical/empirical, qual/quant).
- Empirical research as exploiting **variation**; research design as inference.
- A few questions on variation — the Socratic block on why we care about variation, the “few observations” critique, why statistics at all, where the variation is in attribution science. Keep all of it.
- Unit of analysis, including What's the unit of analyses in the data behind this graph?
- Not only about data or empirical evidence — the point that theses fail on the non-data part. This is the course's thesis statement; move it earlier.
- Logistics, calendar, textbooks, assessment.

#### *Cut / compress*

- Research process in detail (1)–(6) → **2 frames**. Six is three too many for week 1; students have no hooks to hang the detail on yet. The detail returns naturally in weeks 2–5.
- Empirical Research and Empirical research (two near-identical titles) → merge.
- Fix Key ingredientes typo.

#### *Add*

- **The triad**: describe / explain / predict, one frame defining it, one frame per goal with its CSS anchor example (Chetty atlas / Guess experiment / Blumenstock phone metadata). ~4 frames. This is the single most important addition in the course.
- **The five design questions** slide — the course map. 1 frame, reused every week.
- **Where does data come from?** first pass: *designed* (survey, experiment, coding scheme) vs. *found* (traces, admin records, scraped text). One frame, plants the flag for session 2.
- **“Audit a claim” exercise** (~15 min, 3–4 frames): three claims on screen, one of them AI/CSS-flavoured, one descriptive, one causal. Students say in pairs: what's the goal,

what would the data have to look like, what's the first thing you'd doubt. Gets them talking in hour one, which sets the participation norm for a small cohort.

### *Milestone briefed at the end*

M1: one sentence naming a topic. Due at the start of session 2.

### *Reading*

Hannah Fry, *What Data Can't Do* (keep). Add Salganik, *Bit by Bit* ch. 1 — free online, short, and it frames the whole MA, not just this course.

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## Session 2 — Questions, and where data comes from (Tue Sep 22) — rebuilt

Source: front half of 2\_basics\_quantitative\_data/basics\_quant\_data.tex + substantial new material. This is the session that changes most.

### *Block A — from topic to answerable question (~25 frames, mostly existing)*

- The RQ ladder, both versions, including the honest “anecdotal argument” step. Keep as-is.
- Good RQs, in brief and Good RQs, more in detail with the three-way counterexample contrast. Keep as-is.
- Example on generating RQs (the school peers vs. teaching quality argument). Keep — it is a good live exercise.
- Stories, RQs, and theories / Generating theories — the abstraction ladder, and the “if you can't tell a story out of the theory you're not there yet” advice. Keep.
- Mark the **Charlie Kirk running example** as a swappable slot; replace with whatever is current in Sep 2026, keeping the same RQ1/RQ2/RQ3 decomposition structure.
- Move here from session 3's current deck: **levels of explanation (macro/meso/micro)** — it belongs with theory-building, not with measurement.

### *Block B — the data-generating process, socially and technically (~28 frames, new)*

This is the CSS core of the course. Sequence:

1. **Designed vs. found data.** Two frames. Designed data answers a question you had; found data answers a question someone else had, or no question at all. Everything else in the block follows from that.
2. **Salganik's ten characteristics of big data sources.** Big / always-on / non-reactive (the attractions); incomplete / inaccessible / non-representative / drifting / algorithmically confounded / dirty / sensitive (the design problems). ~6 frames. This is a *research design* framework with no statistics in it, and it gives students a checklist they will use for the rest of the MA.
3. **Non-representative by construction** — the Xbox survey (Wang et al. 2015). An absurdly unrepresentative panel producing good election forecasts. The lesson is that representativeness is a property of the **design**, not of the sample. ~3 frames.

4. **Drift** — Google Flu Trends (Lazer et al. 2014). A measure that silently stops meaning what it meant, partly because the platform changed underneath it. ~4 frames. Doubles as the session's assigned paper.
5. **Algorithmic confounding** — the platform is in your DAG. Ranking and recommendation shape the behaviour you observe, so an observed association can be a property of the system rather than of the people. ~3 frames. Forward-references DAGs in session 4.
6. **Access and the post-API age** (Freelon 2018) — what happens to a design when the data source can revoke access, and what that means for reproducibility. ~2 frames. Practical: several of them will propose scraping something for the final essay.
7. **Ethics preview** — one frame, informational risk and consent at scale, pointing to the full treatment in session 5. Flag it early because essay proposals get written in week 4.

### Milestone

M1 collected. M2 briefed: topic → question, plus units and the variation you'd exploit.

### Reading

Lazer, Kennedy, King & Vespignani (2014), *The Parable of Google Flu*, Science. Three pages, no math, and it is simultaneously about measurement validity, drift, and big-data hubris. (Müller-Crepon moves to session 3.)

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## Session 3 — Do the numbers mean what you think? (Tue Sep 29) — promoted to a full session

Source: back half of `basics_quant_data.tex` (which already has ~40 frames of good material here), plus new CSS measurement content.

### Block A — concepts and measurement (~28 frames, mostly existing)

- Concepts as building blocks; rule-based vs. ideal types; the household exercise; the political-violence exercise. Keep.
- Operationalization, with *operationalize* ≠ *measure*. Keep.
- Importance — the point that a huge share of good quantitative work *is* conceptual and operational improvement, and that this matters **especially** for CSS. Keep and sharpen.
- Civil war / outbreak worked example. Keep.
- The three measurement issues, proxies, latent variables, validation. Keep.
- Democratic backsliding: objective indicators vs. V-Dem. Keep — best measurement frame in the course.
- Missing data and sampling bias as measurement problems. Keep.

### Block A additions (new, ~8 frames)

- **A classifier's output as a variable.** When your independent variable is the output of a model — a topic label, a sentiment score, an estimated ideology — the measurement question doesn't go away, it gets harder, and the error is *not* random. Use Barberá's Twitter ideal points (images already in `img/barbera_tw*.jpg`, `img/barbera_eta1*.jpg`) as the well-done

case: what was validated against what. Then the general rule from Text-as-Data: validate, always, against something external.

- **A proxy that encodes the thing you didn't mean** — Obermeyer et al. (2019): an algorithm allocating care used *healthcare cost* as a proxy for *health need*. Purely a construct validity failure, no statistics needed to see it, and it lands hard with this cohort. ~3 frames.

### *Block B — description as a first-class goal (~18 frames, existing + expanded)*

The current deck has this material but it sits at the end of an overloaded session and gets cut. Given a proper slot: - Describing variables, describing relationships, univariate and bivariate. Keep. - Is description useful? — expand from a rhetorical aside into the actual argument. Description has its own design requirements: coverage, comparability, a defensible denominator, and a clear population. - **Add:** Chetty et al. (2022) social capital — a purely descriptive project at enormous scale, with real design decisions and real consequences. The example that proves description isn't the consolation prize. - **Add:** culturomics (Michel et al. 2011) and its critique (Pechenick et al. 2015) — a descriptive claim about culture that turns out to be partly a claim about what got digitised. Ties back to session 2's drift and to corpus composition. ~3 frames. - Wartime civilian deaths worked example. Keep as the closing synthesis.

### *Milestone*

M2 collected. M3 briefed: your key concept, how you'd measure it, one threat to that measure.

### *Reading*

Müller-Crepon, Hunziker & Cederman, *Roads to Rule, Roads to Rebel* (moved here from session 2). It is fundamentally an operationalization paper — a relational concept of state capacity built out of road networks — so it belongs in the measurement session, where the discussion can be “what concept, what operationalization, what would you have done instead.”

---

## **Session 4 — What comparison licenses a causal claim? (Tue Oct 6)**

Source: 3\_causality/causality.tex (91 frames → trim to ~55).

### *Keep*

- Explaining relationships; potential outcomes; the fundamental problem.
- The PO worked example — but **compress from 8 frames to ~5**.
- ATE vs. ATT, including why ATT is often the useful one in practice.
- Experiments and their threats: randomization failure, SUTVA, attrition, compliance, external validity. Keep all five.
- DAGs: what one is, mediation vs. moderation, the income-inequality example, DAGs and mechanisms.
- Front doors and back doors, the full ten-frame build. Keep — it earns the length.
- **Usual suspects:** confounding, reverse causality, bidirectional causation, selection bias, collider bias (with the animation), post-treatment bias, and the closing what should not be controlled for recap. Keep entirely. Best block in the course.

*Cut / compress*

- Off topic: Controlling  $\times 4 \rightarrow$  **1 conceptual frame**. Controlling as blocking a back door, not as adding a regressor. The regression version belongs to AQM.
- Restore the **prediction** frames here or in session 1 (currently 6 commented frames). If the triad is properly introduced in session 1, one or two frames here suffice: prediction needs no causal identification, which is exactly why it can't answer a causal question. Hofman, Sharma & Watts (2017) is the one-page reference.

*Add (new, ~6 frames)*

- **Platform experiments as the modern experimental ideal** — and their limits. Guess et al. (this week's paper) as the well-resourced case: real randomization, huge N, and *still* bounded external validity because it ran on one platform in one campaign. Pair with Bail et al. (2018), where exposure to opposing views on Twitter *increased* polarization — a clean, memorable experiment whose finding contradicts the intuitive mechanism.
- **Interference is the normal case in CSS, not an edge case**. SUTVA is introduced above as an assumption; make the CSS point explicitly, because it is where a lot of student designs on network or platform data quietly fail. If treating one user affects their contacts, there is no clean control group, and the design has to be built around that from the start rather than patched afterwards. No math required to make this land.

*Milestone*

M3 collected. M4 briefed: the comparison you'd make, and what would bias it.

*Reading*

Guess et al. (2023), *How do social media feed algorithms affect attitudes and behavior in an election campaign?* Science. Keep — and it now sits in the right session.

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## **Session 5 — What variation can you exploit, and how far does it travel? (Tue Oct 13) — rebuilt**

Source: 4\_causal\_inference\_methods/causal\_inference.tex (58 frames, well-calibrated) + the first ~45 frames of 5\_advanced\_topics/advanced.tex. The remaining ~65 frames of advanced.tex are cut.

*Block A — the design templates, at design level only (~28 frames)*

- **Exogenous variation** as the unifying idea. Keep the existing framing — it is already the right one.
- Overview of the 3 design templates. Promote this to the organizing frame of the block.
- For each of **matching/controlling, fixed effects, DiD, RDD, IV**, exactly two things: *what variation does it exploit*, and *what has to be true for it to work*. Roughly 3–4 frames each, no estimation. Existing frames already do this well in this deck; the job is resisting the pull toward the detailed versions.

- Keep Snow’s cholera (img/did\_snow.pdf), the street-names DiD (img/did\_TJ\*), and the IV and RDD worked examples.
- Keep Alternative approaches to IV: build your own — good for essay proposals.
- Keep Controlling and exploiting exogeneity — the point that designs combine.

### *Block B — generalization and credibility (~25 frames)*

- **External validity** (Egami & Hartman 2023) and **construct validity**. Keep.
- **Meta-analysis as a design idea** — the natural-resources example. Compress from ~10 frames to ~5: the point is that a literature is itself an object of study, and that funnel plots reveal what didn’t get published.
- **Temporal validity** (Munger 2023). Keep — and it is especially pointed for CSS, where the platform under study may not exist in five years.
- **Placebo and robustness tests as design logic**, not as statistics: what would have to be false for your result to be an artifact, and can you go look. Keep.
- Synthetic control — keep as **appendix**, cover only if time.

### *Block B additions (new, ~10 frames)*

- **Researcher degrees of freedom and the forking paths problem**. Then Breznau et al. (2022): 162 researchers, same data, same hypothesis, results spanning both signs. Nothing in it requires understanding a model, and it is the strongest possible argument for why you commit to a design *before* seeing results. ~4 frames.
- **Pre-registration and pre-analysis plans** — and the explicit statement that the final essay is a pre-analysis plan. This reframes the assignment from “an essay about methods” to “the actual document a researcher writes,” which noticeably improves what students hand in. ~2 frames.
- **Research ethics for computational work** — informational risk, consent at scale, identifiability of “anonymous” data. Emotional Contagion (Kramer et al. 2014), the Tastes/Ties re-identification, Netflix de-anonymization, and GDPR as the framework these students will actually be bound by. ~4 frames. There may be nowhere else in the MA where this is covered, and students proposing to scrape or link data need it *before* they write their essays, not after.
- **Closing: prediction revisited**. The Fragile Families Challenge (Salganik et al. 2020) — hundreds of teams, unusually rich longitudinal data, best models still predicting life outcomes poorly. Return to the session-1 triad, close the course on calibrated humility rather than on a list of methods. ~3 frames.

### *Cut from this session*

The entire Causal methods again, in detail section of advanced.tex — ~65 frames covering controlling in regression, matching mechanics, propensity scores, FE ( $\times 13$ ), DiD ( $\times 13$ ), RDD ( $\times 10$ ), IV ( $\times 8$ ). Duplicative of Block A and outside the course’s stated scope. Move to slides/appendix\_methods\_detail/ and offer as optional self-study for students continuing to AQM.

### **Milestone**

M4 collected. M5 briefed: 1-page design sketch, swapped with an assigned peer before the workshop.

### **Reading**

None assigned — protect the time for questions before the essay. Optional: Salganik, *Bit by Bit* ch. 6 (ethics).

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## **Session 6 — Workshop (Mon Oct 26, 15h–21h)**

Structure is currently “12–15 slots, details in class.” Proposed changes in 06\_assessment.md; the substantive ones:

- Group presentations must include a slide titled “**the threat I can’t solve.**” Removes the incentive to oversell, and produces much better discussion.
- Presenters must state **which of the three goals** their project has (describe / explain / predict) and **which design template** they’re using, in the first minute. Closes the loop on the spine.
- Peer reviewers are **assigned in advance** and have read the M5 sketch. Reviewing cold in the room produces vague feedback; reviewing a document produces specific feedback.
- Timeboxing for ~12–15 slots across 6h: 12 min present, 8 min discussion, hard stop.

## Examples catalogue

Organised by what each one *teaches*, since the constraint is design lessons per minute, not coverage. Marked **[have]** where images already exist in slides/img/, **[new]** where assets need sourcing.

### Already in the course — keep

| Example                                     | Session | What it teaches                                                                          | Assets                                                                                |
|---------------------------------------------|---------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Challenger / O-rings                        | 1       | Selection on the dependent variable; the answer is in the data you <i>didn't</i> look at | <b>[have]</b> challenger1.jpg, challenger2.jpg, challenger_wiki.jpg                   |
| Survivorship bias (planes)                  | 1       | Same lesson, one image                                                                   | <b>[have]</b> files/survivorship_bias.png                                             |
| Hotel bookings vs. visits                   | 1       | Measuring the wrong thing because it's the thing you have                                | —                                                                                     |
| Attribution science                         | 1       | Where is the variation when there's one case?                                            | —                                                                                     |
| Barberá Twitter ideal points                | 3       | Latent measurement from digital traces, done carefully and validated                     | <b>[have]</b> barbera_tw1-4.jpg, barbera_etal1-2.jpg                                  |
| Democratic backsliding: objective vs. V-Dem | 3       | Same concept, two operationalizations, different answers                                 | <b>[have]</b> democratic_backsliding.jpg, economist_backsliding.png                   |
| Wartime civilian deaths                     | 3       | End-to-end synthesis: concept → operationalization → measure → description               | <b>[have]</b> battledeaths.png, declineofwar.png                                      |
| Müller-Crepon roads                         | 3       | Building a relational concept out of spatial data                                        | <b>[have]</b> mullercrepon.png, carl1.png, carl9.png, carl10.png, guatemala_roads.png |
| Collider / M-bias sequence                  | 4       | The one bias nobody's intuition gets right                                               | <b>[have]</b> collider1-5.pdf, collider_animation.gif, collider_m1-2.png              |
| Guess et al. feed experiment                | 4       | Platform experiment: what randomization buys, what it doesn't                            | — (paper)                                                                             |
| Snow's cholera                              | 5       | The original natural experiment                                                          | <b>[have]</b> did_snow.pdf                                                            |
| Street names / TJ backlash                  | 5       | DiD from the instructor's own work; students see the whole design                        | <b>[have]</b> did_TJ*, map_graves*.pdf                                                |
| Natural resources meta-analysis             | 5       | A literature as an object of study; funnel plots                                         | <b>[have]</b> existing frames                                                         |
| Blumenstock phone metadata                  | 1, 3    | Prediction from found data; also a measurement story                                     | <b>[have]</b> blumenstock.png, blumenstock2.png, blumenstock3.png                     |

### To add — highest value first

#### 1. Google Flu Trends — Lazer, Kennedy, King & Vespignani (2014), Science

**Session 2. Also the assigned reading.** Three pages, no equations. Teaches drift, big-data hubris, and the fact that a measure can silently stop meaning what it meant — partly because the platform changed underneath it. The single best CSS design cautionary tale, and it works as both lecture example and discussion paper. **[new]** — the CDC vs. GFT overprediction time series plot.

## 2. Xbox election forecast — Wang, Rothschild, Goel & Gelman (2015), IJF

**Session 2.** A panel of Xbox gamers — overwhelmingly young, overwhelmingly male — producing good 2012 forecasts. Teaches that **representativeness is a property of the design, not the sample**, which is the correct antidote to “my Twitter data isn’t a random sample so it’s useless” and to “I have 10 million observations so it doesn’t matter.” Rare example that cuts both ways. **[new]** — the demographic-skew bar chart and the forecast-vs-outcome plot.

## 3. Fragile Families Challenge — Salganik et al. (2020), PNAS

**Session 5, closing.** 160 teams, 12,942 predictor variables, thousands of hours, and the best models still predicted life outcomes poorly. Teaches the limits of prediction with unusually rich data, and closes the course on the triad from session 1. Ends on calibrated humility instead of a method list, which is the right note for week 1 of a master’s. **[new]** — the scatter of submitted models’  $R^2$  against the benchmark.

## 4. Many analysts, one dataset — Breznau et al. (2022), PNAS

**Session 5.** 162 researchers, same data, same hypothesis, estimates spanning both signs. The most persuasive available argument for committing to a design before seeing results, and it requires understanding no model at all — just the spread of the dots. **[new]** — the specification-curve / distribution-of-estimates figure.

## 5. Algorithmic proxy failure — Obermeyer et al. (2019), Science

**Session 3.** An algorithm allocating extra care used *healthcare spending* as a proxy for *health need*. Since less was historically spent on Black patients at equal sickness, the proxy encoded the disparity. Pure construct validity, visible without any statistics, and it connects measurement to consequences in a way this cohort will remember. **[new]** — the two-panel cost-vs-illness-by-race figure.

## 6. Bail et al. (2018), PNAS — bots exposing users to opposing views

**Session 4.** A real field experiment on a platform whose result runs *against* the intuitive mechanism: exposure to the other side increased polarization, especially among Republicans. Excellent paired with Guess et al. for a discussion of mechanism vs. effect, and of why a well-identified estimate can still surprise you. **[new]** — the treatment-effect-by-party figure.

## 7. Chetty et al. (2022), Nature — social capital

**Session 3.** 21 billion Facebook friendships turned into a descriptive atlas of cross-class connectedness. Proves description is a first-class goal with real design decisions: what’s the unit, what’s the denominator, what population. Also a data-access and privacy story worth one sentence. **[new]** — one Opportunity Insights map.

## 8. Culturomics and its critique — Michel et al. (2011) / Pechenick et al. (2015)

**Session 3.** A descriptive claim about culture that turns out to be partly a claim about what got digitised — the corpus is heavily shaped by scientific publishing over time. Ties back to session

2's drift, and teaches corpus composition as a measurement problem. **[new]** — an n-gram trend plus the corpus-composition-over-time figure.

### 9. MusicLab — Salganik, Dodds & Watts (2006), Science

**Session 4, optional.** Parallel artificial music markets: the same songs become hits in some worlds and not others. Teaches cumulative advantage, and — more usefully here — that **a designed digital experiment can create the counterfactual worlds observation can't give you**. Good if the experiments block has room. **[new]** — the download-inequality-across-worlds figure.

### 10. Emotional Contagion — Kramer, Guillory & Hancock (2014), PNAS

**Session 5, ethics.** Feed manipulation at scale with no meaningful consent. Pair with the Tastes/Ties re-identification and the Netflix Prize de-anonymization. The discussion question that works: *the analysis was fine — so what exactly was wrong?* Forces students to separate methodological from ethical evaluation. **[new]** — no figure needed; a headline screenshot is enough.

### Spain / Europe-specific options

Worth having one or two so the cohort sees designs on data they could actually get:

- **COVID mobility data (INE mobile-phone study)** — a designed use of found telecom data; good for unit of analysis (what is a “trip”?) and for privacy-by-aggregation. Relevant assets may exist: `img/acumuladaincidencia31marzo.jpeg`, `covid_*.png`.
- **Spanish street names** — already in the course, instructor's own.
- **Spanish electoral data** — `img/electoral_data_excel.jpg` already exists and is a nice “this is what real data actually looks like” moment for week 2 or 3.

### Swappable current-event slot

Session 2's RQ decomposition currently runs on the Charlie Kirk / political violence example. It works, but it dates. Keep the **structure** (topic → intuition → main RQ → RQ1/RQ2/RQ3 → theory) and mark the slot in the .tex with a comment so it's obvious what to replace annually. Whatever is salient in Sep 2026 will do, provided it decomposes into a descriptive, a distributional, and a causal sub-question — that three-way split is the pedagogical point, not the topic.

### Exercises (not examples, but they need slots)

| Exercise                                                                                                 | Session | Time   | What it does                                        |
|----------------------------------------------------------------------------------------------------------|---------|--------|-----------------------------------------------------|
| <b>Audit a claim</b> — 3 claims, pairs, name goal + data + first doubt                                   | 1       | 15 min | Sets participation norm in hour one                 |
| <b>Concept stress-test</b> — define “political violence” / “household” as rule-based, then as ideal type | 3       | 10 min | Already in the deck; keep it live, don't lecture it |
| <b>Find the back door</b> — a DAG on screen, students name what to control and what not to               | 4       | 15 min | The collider lesson only sticks if they try it      |

| Exercise                                                                                            | Session | Time   | What it does                                |
|-----------------------------------------------------------------------------------------------------|---------|--------|---------------------------------------------|
| <b>Name the variation</b> — 4<br>published abstracts, students<br>identify which design<br>template | 5       | 15 min | Best possible rehearsal for the<br>workshop |

## Readings

Constraint: seven sessions, students at the start of a 1-year MA, three graded paper commentaries. Keep **one assigned item per session**, short, and with no statistics the students can't skip past.

### Assigned — one per session

#### Session 1 — Introduction

- Hannah Fry, [“What Data Can't Do”](#), *The New Yorker*, 29/03/2021. **keep** — ungraded, sets tone.
- *Optional*: Matthew Salganik, *Bit by Bit* ch. 1 (“Introduction”). Free online, ~15 pages. Worth assigning because it frames the whole MA.

#### Session 2 — Questions and data provenance — [changed]

- David Lazer, Ryan Kennedy, Gary King & Alessandro Vespignani (2014), “The Parable of Google Flu: Traps in Big Data Analysis,” *Science* 343(6176): 1203–1205.  
**Why this one:** three pages, zero equations, and it carries the whole session — drift, measurement validity, big-data hubris, and the platform as part of the DGP. Graded commentary #1.  
*Discussion prompts:* What was GFT actually measuring? Whose behaviour changed — the public's, or Google's? Would more data have fixed it? What would you have had to do in 2009 to notice this coming?
- *Optional companion:* Deen Freelon (2018), “Computational Research in the Post-API Age,” *Political Communication* 35(4): 665–668. Two pages, and directly relevant to students about to propose scraping something.

#### Session 3 — Measurement and description

- Carl Müller-Crepon, Philipp Hunziker & Lars-Erik Cederman (2021), [“Roads to Rule, Roads to Rebel”](#), *JCR* 65(2–3): 563–590. [**kept, moved from session 2**]  
**Why here:** it is fundamentally an operationalization paper — a relational concept of state capacity constructed out of road networks. In the measurement session the discussion becomes “what concept, what operationalization, what would you have done instead,” which is a much better use of it than a general discussion of quantitative research. Graded commentary #2.  
*Discussion prompts:* What is the concept, in one sentence, without using the word “roads”? What does the road network fail to capture? Tell me a story where their measure goes up and actual state capacity goes down.

#### Session 4 — Causality

- Andrew Guess et al. (2023), [“How do social media feed algorithms affect attitudes and behavior in an election campaign?”](#), *Science* 381(6656): 398–404. **keep**  
Graded commentary #3. Students can skip the statistics entirely and still discuss the design, which is the point.

*Discussion prompts:* What exactly was randomized, and what wasn't? Who consented, and how does that bound the estimate? If the effect is near zero, what have we learned — about feeds, or about three months? Where does SUTVA fail here?

### *Session 5 — Designs, generalization, credibility*

- **No assigned reading.** Protect the time for essay questions; this is the last session before the workshop.
- *Optional:* Salganik, *Bit by Bit* ch. 6 (“Ethics”). Free online.

### **Instructor background — not assigned**

Useful for building the new blocks; too long or too technical for this cohort.

**Data provenance / CSS design (session 2)** - Matthew Salganik (2018), *Bit by Bit: Social Research in the Digital Age*, Princeton UP. Ch. 2 (“Observing Behavior”) is the source for the ten-characteristics framework. Free at bitbybitbook.com. **The single most useful book for this reframing.** - Wang, Rothschild, Goel & Gelman (2015), “Forecasting elections with non-representative polls,” *IJF* 31(3): 980–991. — the Xbox study. - Timnit Gebru et al. (2021), “Datasheets for Datasets,” *CACM* 64(12). — good source for a “what you should be able to say about your data” checklist slide.

**Measurement (session 3)** - Justin Grimmer, Margaret Roberts & Brandon Stewart (2022), *Text as Data*, Princeton UP. Ch. on validation; the “no free lunch” principle. - Ziad Obermeyer et al. (2019), “Dissecting racial bias in an algorithm used to manage the health of populations,” *Science* 366(6464): 447–453. - Dean Knox, Christopher Lucas & Wendy Cho (2022), “Testing Causal Theories with Learned Proxies,” *Annual Review of Political Science* 25. — background for the classifier-output-as-variable slides; the intuition is teachable, the paper isn't. - Raj Chetty et al. (2022), “Social capital I,” *Nature* 608: 108–121. - Michel et al. (2011), “Quantitative Analysis of Culture Using Millions of Digitized Books,” *Science* 331: 176–182; and Pechenick, Danforth & Dodds (2015), “Characterizing the Google Books Corpus,” *PLoS ONE* 10(10).

**Estimands and theory (session 4)** - Ian Lundberg, Rebecca Johnson & Brandon Stewart (2021), “What Is Your Estimand?” *ASR* 86(3): 532–565. **Strong fit for this course's philosophy** — it's about connecting a theoretical question to a defined target quantity, and most of it needs no statistics. Consider assigning excerpts in a future year, or lifting its central diagnostic (“state your estimand in words”) as an essay requirement. - Christopher Bail et al. (2018), “Exposure to opposing views on social media can increase political polarization,” *PNAS* 115(37): 9216–9221. - Nyhan et al. (2023), “Like-minded sources on Facebook are prevalent but not polarizing,” *Nature* 620. — companion to Guess et al. from the same research programme.

**Prediction (sessions 1 and 5)** - Jake Hofman, Amit Sharma & Duncan Watts (2017), “Prediction and explanation in social systems,” *Science* 355(6324): 486–488. Two pages; the cleanest statement of the distinction. - Jake Hofman et al. (2021), “Integrating explanation and prediction in computational social science,” *Nature* 595: 181–188. - Matthew Salganik et al. (2020), “Measuring the predictability of life outcomes with a scientific mass collaboration,” *PNAS* 117(15): 8398–8403. — Fragile Families Challenge. - Kleinberg, Ludwig, Mullainathan & Obermeyer (2015), “Prediction Policy Problems,” *AER P&P* 105(5): 491–495.

**Credibility and ethics (session 5)** - Nate Breznau et al. (2022), “Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty,” *PNAS* 119(44). -

Kramer, Guillory & Hancock (2014), “Experimental evidence of massive-scale emotional contagion through social networks,” *PNAS* 111(24): 8788–8790. - Michael Zimmer (2010), “‘But the data is already public’: on the ethics of research in Facebook,” *Ethics and Information Technology* 12(4). — the Tastes/Ties case. - Egami & Hartman (2023), “Elements of External Validity,” *APSR* 117(3): 1070–1088. [already cited in deck 5] - Kevin Munger (2023), “Temporal Validity as Meta-Science,” *Research & Politics*. [already cited in deck 5]

### Textbook list on the syllabus

Current list (Huntington-Klein, Imai, Toshkov, Cunningham) is fine and should stay. One addition is worth making explicit for this cohort:

- **Salganik, *Bit by Bit*** — free online, no statistics, and it is the only one of these books that is *about* computational social science research design rather than about estimation. Given the MA’s focus it arguably belongs first on the list.

Note that Imai and Cunningham are both considerably more technical than this course; worth a one-line annotation on the syllabus saying which books are for *this* course and which are for the methods courses that follow. Students at the start of a master’s read a booklist as a set of instructions.

## Assessment and the milestone ladder

### Weights

The comment at the top of `syllabus_research_design.tex` records the intent to raise the final essay to 50–60% and drop participation. Implemented:

| Component                       | Current    | Proposed   |
|---------------------------------|------------|------------|
| Participation                   | 15%        | —          |
| Paper commentaries (3 × 5%)     | 15%        | 15%        |
| Workshop presentation           | 20%        | 20%        |
| Workshop feedback (peer review) | 10%        | 10%        |
| <b>Final essay</b>              | <b>40%</b> | <b>55%</b> |

Dropping participation is the right call for a seven-session course: it is hard to grade defensibly, and the paper commentaries already reward engagement in a way that leaves a paper trail. The expectation of active participation stays in the syllabus text as an expectation, just not as a number.

### The milestone ladder — the main addition

The problem: the essay is 55% and is the first research design most of these students have ever written, and the only checkpoint before the workshop is a recap slide in session 4. In practice students start thinking about their design in week 5.

The fix: five short deliverables, each due at the start of a session, each **directly produced by that week’s material**, and each a paragraph of the eventual essay.

| #         | Due                    | Deliverable                                                                | Length        | Feeds on  |
|-----------|------------------------|----------------------------------------------------------------------------|---------------|-----------|
| <b>M1</b> | Session 2              | A topic, and why anyone should care                                        | 2–3 sentences | Session 1 |
| <b>M2</b> | Session 3              | Topic → research question. Name the units, and the variation you’d exploit | ~half page    | Session 2 |
| <b>M3</b> | Session 4              | Your key concept, how you’d measure it, and one threat to that measure     | ~half page    | Session 3 |
| <b>M4</b> | Session 5              | The comparison you’d make, and what would bias it                          | ~half page    | Session 4 |
| <b>M5</b> | 3 days before workshop | 1-page design sketch, assembled from M1–M4                                 | 1 page        | Session 5 |

Design notes:

- **Ungraded, but required.** Checked as done / not done, folded into the essay grade only in the sense that a student who did them writes a better essay. This keeps grading cost near zero — five items × 12 students is about 40 minutes a week of skimming, and most of it is reading three sentences.
- **Feedback is oral and collective**, not written. Open each session by putting two or three anonymised M-submissions on screen and workshopping them for ten minutes. This is the highest-value teaching in the course: students learn far more from watching a real question

get sharpened than from a lecture about sharpening questions. It also costs no prep beyond choosing which ones.

- **M5 goes to the assigned peer reviewer**, not just to the instructor. Peer reviewers who have read a document give specific feedback; reviewers hearing a talk could give vague feedback. This is what makes the 10% peer-review component gradeable.
- **The essay is explicitly a pre-analysis plan.** Say this in session 1, repeat it in session 5 when pre-registration is covered. It converts a vague “essay about research design” into a document with a known genre, and it is genuinely what the assignment is.

Each milestone maps to one of the five design questions from `02_spine.md`, so the ladder and the course structure are the same object seen twice. That is deliberate: by session 5 students have answered all five questions about their own project, and the essay is assembly plus prose.

### Final essay — sharpen the prompt

The current prompt (a pre-analysis plan or an overview of how to explore a topic quantitatively, with the structure suggested in the syllabus) is good. Three additions:

1. **State the research goal explicitly.** The essay must say, in the first paragraph, whether it is descriptive, explanatory, or predictive — and the design must match. This closes the loop on the triad, and it heads off the most common failure: a descriptive question with a causal answer bolted on.
2. **Require a “threats” section** with at least one threat the student cannot solve. Currently the incentive is to present a design as airtight; a design with no acknowledged limitations is either trivial or not understood. This also matches what the workshop asks for.
3. **Require a data provenance paragraph.** Where does the data come from, who created it and why, what is missing from it, and can you actually get it. For a CSS cohort proposing to use found data, this is the paragraph most likely to save them from an unworkable thesis.

Keep group essays allowed. Keep the deadline structure (roughly a week after the workshop) — it lets students incorporate feedback, which is the whole point of holding a workshop.

### Paper commentaries

Keep as-is: no summary, a few discussion points, brought or sent before class, 5% each. The one change is which papers, per `05_readings.md` — the commentaries now attach to Google Flu (session 2), Müller-Crepon (session 3), and Guess et al. (session 4).

Worth adding to the syllabus: a one-line statement of what a good commentary looks like, since this cohort has likely never written one. Something like *“three or four specific points you would raise if the author were in the room; at least one about the design rather than the findings.”*

### Workshop format

6 hours, ~12–15 slots. Proposed:

- **12 min present / 8 min discuss / hard stop.** 15 slots  $\times$  20 min = 5h, leaving an hour for breaks and a closing session.
- **Required slides:** (1) the goal — describe/explain/predict; (2) the question; (3) units and variation; (4) the concept and its measure; (5) the comparison; (6) **the threat I can’t solve.**

- **Assigned peer reviewers**, one or two per presenting group, who have read the M5 sketch and come with two written comments. Hand these in — that is the 10%.
- **Close with 20 minutes on what the room got wrong collectively.** After 15 presentations there will be two or three recurring errors, and naming them is worth more than any single piece of individual feedback.

## Triaged task list

**Today: 2026-09-03. Session 1 is Tue Sep 15 — 12 days.** Sessions 2–5 fall on Sep 22, Sep 29, Oct 6, Oct 13; workshop Mon Oct 26.

The full reframing in `03_sessions.md` is roughly 2–3 days of concentrated work. That is probably not available before Sep 15, but it doesn't need to be: only session 1 has to be ready on Sep 15, and the sessions that change most (2, 3, 5) are 19, 26 and 40 days out. The sequencing below exploits that.

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### Before Sep 15 — must do (≈half a day)

- ☐ **Update the syllabus.** `syllabi/current/syllabus_research_design.tex` still says *Fall 2025* and Sep 16; `index.md` already has the 2026 dates. Fix the term, the dates, and the session table.
- ☐ **Set the new assessment weights** in the syllabus (essay 55%, participation dropped) — per the instructor note already at the top of that file. This has to be right on day one because students are told the weights in session 1.
- ☐ **Add the milestone ladder to the syllabus** (`06_assessment.md`) — also has to be announced in session 1, and it's five table rows.
- ☐ **Swap the session 2 reading** to Lazer et al. (Google Flu) and move Müller-Crepon to session 3, in both the syllabus and `index.md`.
- ☐ **Session 1 deck edits** — the only deck needed on Sep 15:
  - add the **triad** frames (describe / explain / predict + one anchor example each) — ~4
  - add the **five design questions** course-map frame — 1
  - add **designed vs. found data** — 1
  - add the **audit a claim** exercise — 3–4
  - compress Research process in detail (1)–(6) to 2
  - merge the two Empirical Research frames; fix Key ingredients
  - update the calendar and assessment frames to the new weights and dates
- ☐ **Update the session-outline text** in the syllabus for sessions 2–5 to match the new titles and content. Students will hold you to this document, so it's worth getting the new framing in there now even if the decks come later.

### Before Sep 22 (session 2) — the big build (≈1 day)

Session 2 is the most-changed session. Nine days after session 1.

- ☐ Split `2_basics_quantitative_data/` into two lecture folders:
  - `2_questions_and_data/` — RQ/theory front half + the new provenance block
  - `3_measurement/` — concepts/measurement/description back half
  - Renumber `3_causality` → `4_causality`, `4_causal_inference_methods` + `5_advanced_topics` → `5_designs_and_credibility`.
- ☐ Move **levels of explanation (macro/meso/micro)** from the measurement material into the theory block.
- ☐ Build the **data provenance block** (~28 frames) per `03_sessions.md`: designed vs. found →

Salganik's ten characteristics → Xbox → Google Flu → algorithmic confounding → post-API access → ethics preview.

- ☐ Source figures for **Google Flu** and **Xbox** (see 04\_examples.md).
- ☐ Replace or refresh the **Charlie Kirk** running example; add a % SWAPPABLE: comment marking the slot for future years.

### Before Sep 29 (session 3) — measurement session (≈half a day)

- ☐ Assemble 3\_measurement/ from the existing back-half frames — most of this already exists and just needs to stop being rushed.
- ☐ Add the **classifier-output-as-a-variable** frames, using the Barberá images already in img/.
- ☐ Add **Obermeyer et al.** proxy-failure frames; source the figure.
- ☐ Expand the **description** block: add Chetty social capital and culturomics + its critique; source figures.
- ☐ Keep the wartime civilian deaths example as the closing synthesis.

### Before Oct 6 (session 4) — causality trim (≈2 hours)

Mostly deletion, so this is the cheapest session to prepare.

- ☐ Trim causality.tex from 91 → ~55 frames: compress the PO example 8 → 5, Off topic: Controlling 4 → 1, move overflow to \section{Appendix}.
- ☐ Restore the commented-out **prediction** frames (one or two, since the triad now carries the load from session 1).
- ☐ Add **platform experiments and their limits** (Guess + Bail) and the **interference-is-normal-in-CSS** frames; source the Bail figure.

### Before Oct 13 (session 5) — the cut and the new block (≈1 day)

- ☐ **Cut the Causal methods again, in detail section** from advanced.tex (~65 frames). Move to slides/appendix\_methods\_detail/ and offer as optional self-study. *Move, don't delete* — it's good material in the wrong course.
- ☐ Merge causal\_inference.tex Block A (design templates, ~28 frames, already well-calibrated) with advanced.tex's first ~45 frames.
- ☐ Compress the meta-analysis block 10 → 5 frames.
- ☐ Build the **credibility block**: forking paths → Breznau many-analysts → pre-registration and "your essay is a pre-analysis plan".
- ☐ Build the **ethics block**: informational risk, consent at scale, Emotional Contagion, re-identification, GDPR.
- ☐ Build the **closing prediction-revisited** frames (Fragile Families) and the return-to-the-triad slide.
- ☐ Source figures for Breznau and Fragile Families.
- ☐ Move synthetic control into the appendix section.

### Before Oct 26 (workshop)

- ☐ Assign peer reviewers and circulate M5 sketches ~3 days ahead.
- ☐ Write the peer-review sheet (two written comments, handed in — this is the 10%).
- ☐ Fix the slot schedule: 12 min / 8 min / hard stop.

- ☐ Prepare the required-slides list, including “**the threat I can’t solve.**”

## Housekeeping, any time

- ☐ `index.md` — update session titles to the new framing; un-comment the slide and reading links as decks are built.
  - ☐ Add `slides/beamer_notes_preamble.tex` and adopt the two-wrapper notes pattern from `CLAUDE.md`. **All five decks currently have zero `\note{}`**. Best done incrementally as each deck is rewritten rather than as a separate project.
  - ☐ Adopt the `% ----` frame-separator convention (per `CLAUDE.md`) in rewritten decks.
  - ☐ Add a one-line annotation to the syllabus textbook list saying which books are for *this* course and which are for the methods courses that follow. Add Salganik.
  - ☐ Consider deleting the LaTeX build artifacts under `slides/` (`.aux`, `.bcf`, `.fls`, `.fdb_latexmk`, `.snm`, `.nav`, `.toc`, `.synctex.gz`) and the stray files `survivorship_bias copy.png` — **ask first**, and check that `.gitignore` covers them going forward.
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## Minimal-change variant

If the full restructuring isn’t happening this year, **three insertions and one deletion** get most of the CSS benefit while leaving the session order and all five decks where they are:

1. **Session 1: add the triad** (describe / explain / predict, ~5 frames). Cheapest change with the largest effect on how students think, and it makes every later session easier to place.
2. **Session 2: insert a 40-minute data provenance block** (~15 frames) — designed vs. found data, Salganik’s ten characteristics, Google Flu, algorithmic confounding. Pay for it by cutting the measurement material that currently gets rushed anyway, and by swapping the reading to Lazer et al. Accept that description stays thin this year.
3. **Session 5: delete the Causal methods again, in detail section** (~65 frames) and spend the freed 60–70 minutes on credibility (Bresnau, pre-registration) and ethics. This is the single highest-value change in the whole plan and it is a *deletion*, so it costs nothing but the decision.
4. **Add the milestone ladder** (`06_assessment.md`). Pure syllabus text, no slides, and it is the thing most likely to improve the essays.

That is realistically one day of work, fits in the 12 days available, and leaves the full restructuring — splitting session 2, promoting measurement to its own session — as next year’s job with this document already written.